

JEE MAIN REVISION MASTER: SEMICONDUCTOR DEVICES AND LOGIC GATES

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Semiconductor Physics: Doping, p-n Junction & Rectifiers	21 PYQs	Device Physics
High	Logic Gates, Truth Tables & De Morgan's Laws	11 PYQs	Digital Electronics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Semiconductor Physics: Doping, p-n Junction & Rectifiers	$n_e n_h = n_i^2$, $f_{\text{ripple, half}} = f_{\text{in}}$, $f_{\text{ripple, full}} = 2 f_{\text{in}}$
Logic Gates, Truth Tables & De Morgan's Laws	$\text{NAND: } Y = \overline{A \cdot B}$, $\text{NOR: } Y = \overline{A + B}$, $\overline{A + B} = \bar{A} \cdot \bar{B}$, $\overline{A \cdot B} = \bar{A} + \bar{B}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Semiconductor Physics: Doping, p-n Junction & Rectifiers (Very High Yield — 21 PYQs)

Core Principle: Doping with pentavalent impurities (As, P, Sb) creates n-type (electrons majority). Doping with trivalent impurities (In, B, Al) creates p-type (holes majority). Mass action law $n_e n_h = n_i^2$. Depletion layer contains immobile uncompensated ions. Forward bias reduces barrier; reverse bias widens barrier. Ideal diode acts as closed switch in forward bias, open switch in reverse bias.

■ **Caution / JEE Pitfall:** In the depletion region of an unbiased p-n junction, there are NO free electrons or holes—only immobile positive and negative donor/acceptor ions!

■ **Speed Trick:** In reverse biased circuits with ideal diodes, replace the reverse-biased diode with an open circuit (break the wire) to simplify the circuit instantly.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A wafer of pure germanium crystal has two parts X and Y. The end X is obtained by doping with arsenic and Y with indium. It is connected to ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "In which of the following figures, diode is reverse biased?..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "An n-type and p-type semiconductor can be obtained by respectively doping pure silicon with..." [Ans: N/A]

3.2 Logic Gates, Truth Tables & De Morgan's Laws (High Yield — 11 PYQs)

Core Principle: Basic gates: AND ($Y = A \cdot B$), OR ($Y = A + B$), NOT ($Y = \bar{A}$). Universal gates: NAND ($Y = \overline{A \cdot B}$) and NOR ($Y = \overline{A + B}$). De Morgan's laws: $\overline{A + B} = \bar{A} \cdot \bar{B}$ and $\overline{A \cdot B} = \bar{A} + \bar{B}$.

■ **Caution / JEE Pitfall:** NAND with all inputs tied together acts as a NOT gate. NOR with all inputs tied together also acts as a NOT gate.

■ **Speed Trick:** KCET Speed Trick: To identify mystery gate circuits, test inputs ($A=0, B=0$) and ($A=1, B=1$) first. This eliminates 2 out of 4 options in under 10 seconds.

Representative Source PYQs:

- [2024 Shift] Q#3 (MCQ): "The upper level of valence band and lower level of conduction band overlap in the case of..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "A truth table for the given circuit is..." [Ans: C]
- [2022 Shift] Q#3 (MCQ): "Which logic gate is represented by the following combination of logic gates?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A wafer of pure germanium crystal has two parts X and Y. The end X is obtained by doping with arsenic and Y with indium. It is connected to a battery as shown in the figure. Which of the following statements is correct?

- (A) X is p-type, Y is n-type and the junction is forward biased
- (B) X is n-type, Y is p-type and the junction is forward biased
- (C) X is p-type, Y is n-type and the junction is reverse biased
- (D) X is n-type, Y is p-type and the junction is reverse biased

Correct Answer: None

Solution Analysis: Part X is doped with arsenic. Arsenic is a Group 15 element, so it has 5 valence electrons. When it is added to germanium, it gives extra free electrons. So, part X becomes an n-type semiconductor. Part Y is doped with indium. Indium is a Group 13 element, so it has 3 valence electrons. When it is added to germanium, it creates holes. So, part Y becomes a p-type semiconductor. In the battery symbol, the longer line represents the positive terminal, and the shorter line represents the negative terminal. From the figure, part X is connected to the positive terminal and part Y is connected to the

Benchmark Problem 2: KCET 2024 (Q#3)

The upper level of valence band and lower level of conduction band overlap in the case of

- (A) silicon
- (B) copper
- (C) carbon
- (D) germanium

Correct Answer: None

Solution Analysis: There is no forbidden energy gap between conduction band and valence band for a conductor (copper). Hence, in copper, valence band and conduction band are overlapped.